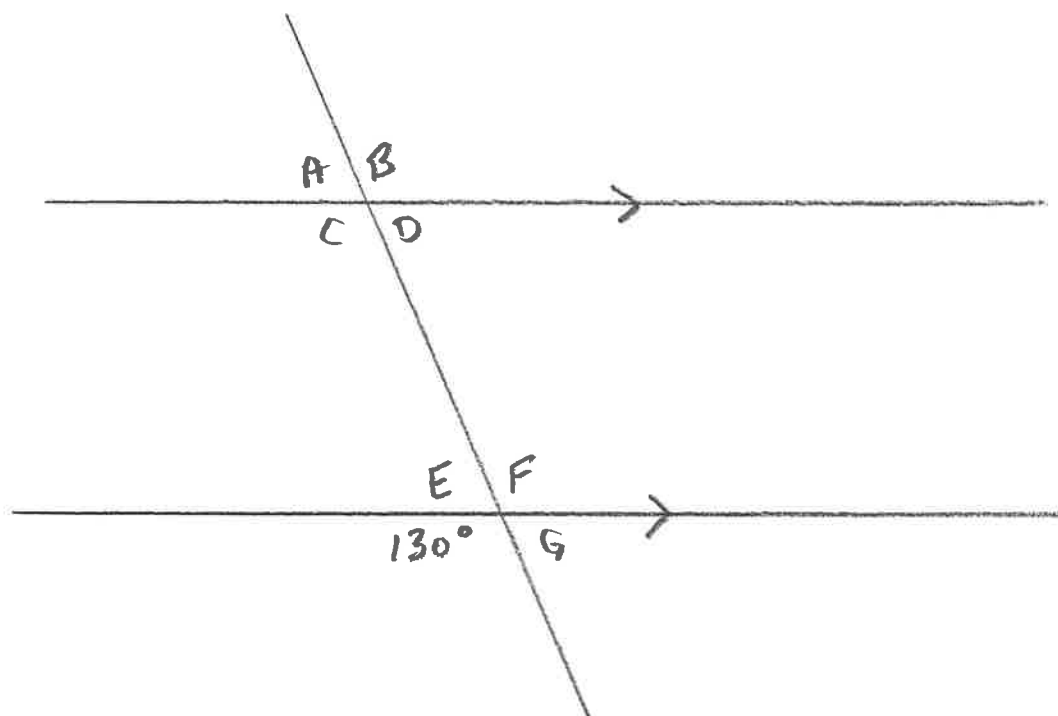


GED Math Assignment 5

#1. See graph paper.

2.



$$\begin{aligned}\angle G &= 180^\circ - 130^\circ \\ &= 50^\circ \text{ (supplementary angles)}\end{aligned}$$

$$\angle E = 50^\circ \text{ (vertical angle with } \angle G)$$

$$\angle F = 130^\circ \text{ (supplementary angle with } \angle E)$$

$$\angle C = 130^\circ \text{ (corresponding angle with } 130^\circ \text{ given angle)}$$

$$\angle A = 50^\circ \text{ (supplementary angle with } \angle C)$$

$$\angle B = 130^\circ \text{ (vertical angle with } \angle C)$$

$$\angle D = 50^\circ \text{ (corresponding angle with } \angle G)$$

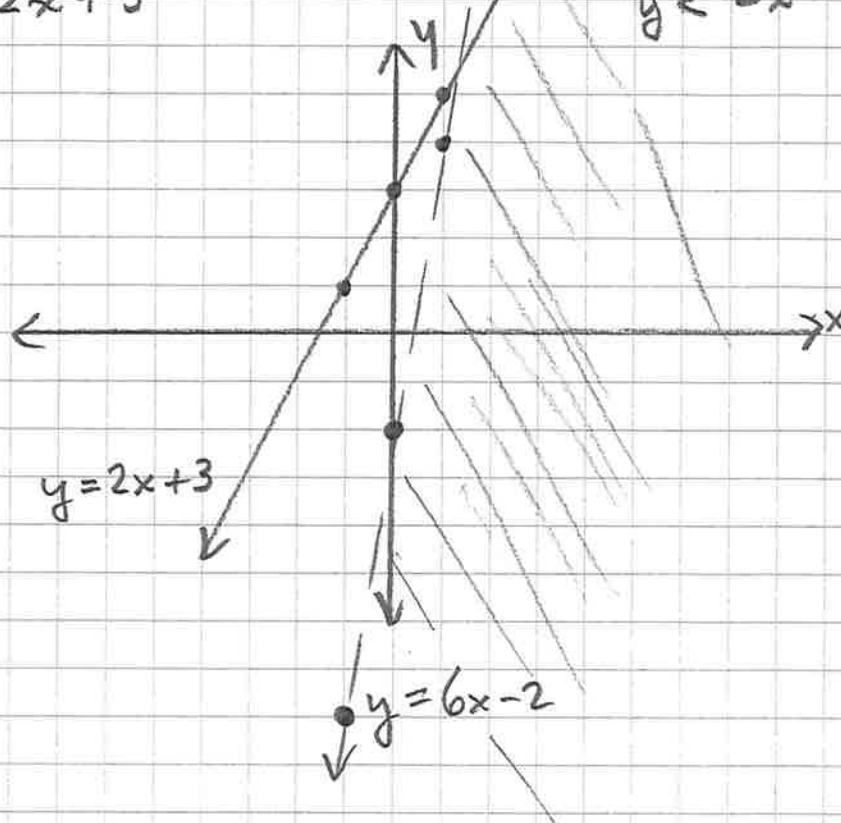
1. (a) $2y - 4x \leq 6$ and $y - 6x \leq -2$

$$2y \leq 6 + 4x$$

$$y \leq 2x + 3$$

$$y \leq -2 + 6x$$

$$y \leq 6x - 2$$



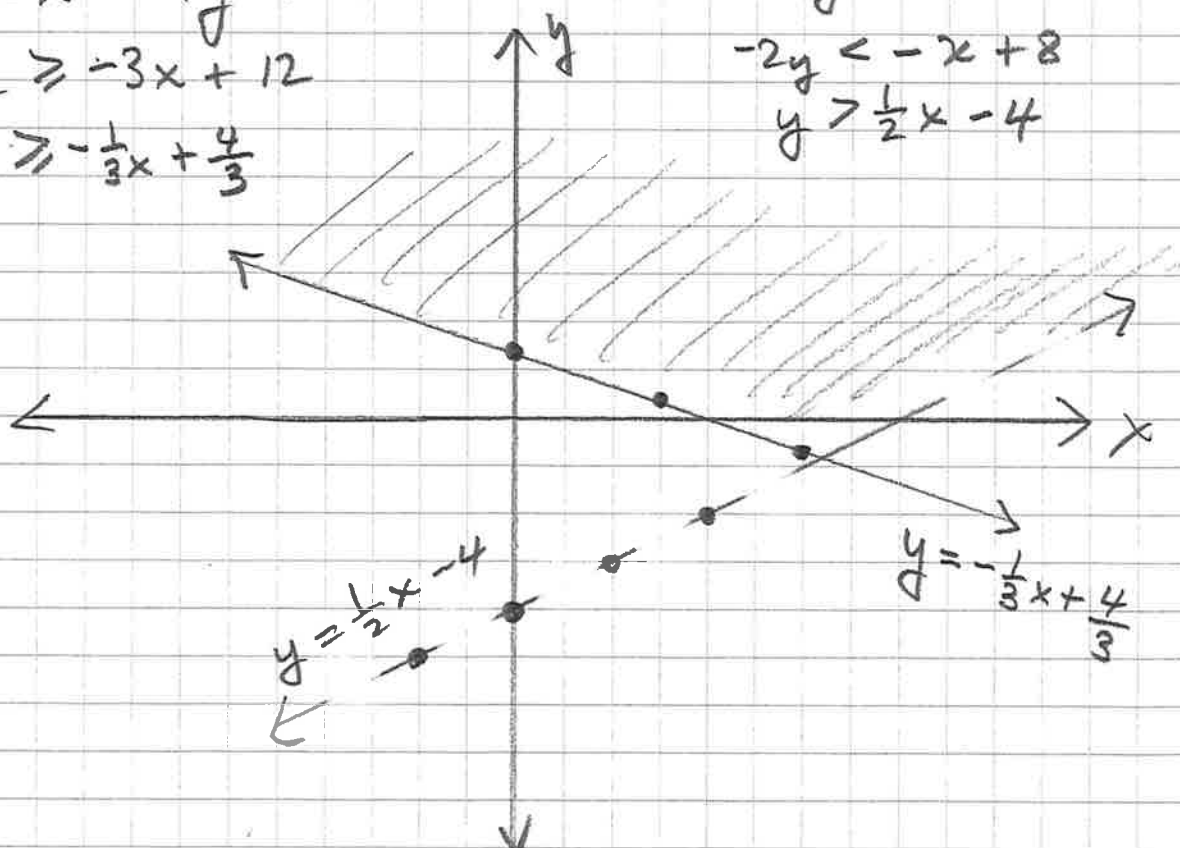
(b) $3x + 9y \geq 12$ and $x - 2y \leq 8$

$$9y \geq -3x + 12$$

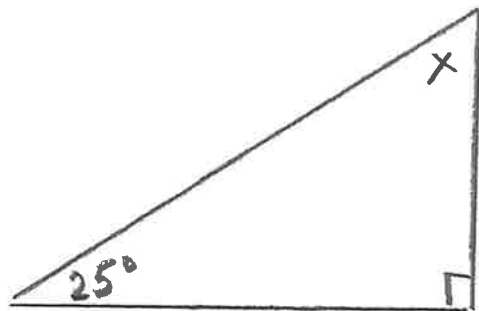
$$y \geq -\frac{1}{3}x + \frac{4}{3}$$

$$-2y \leq -x + 8$$

$$y \geq \frac{1}{2}x - 4$$

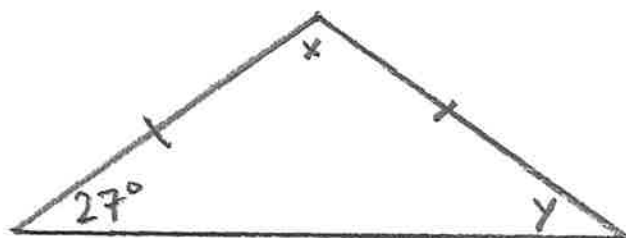


3. (a)



$$\begin{aligned}\angle x &= 180^\circ - 90^\circ - 25^\circ \\ &= 65^\circ\end{aligned}$$

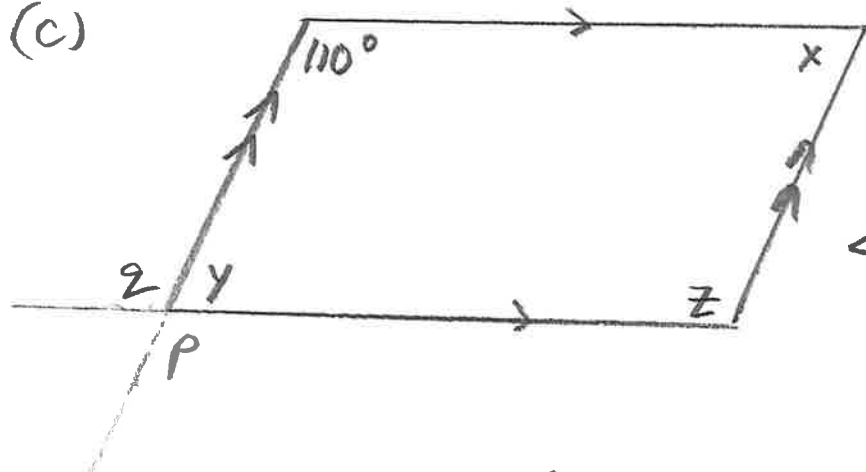
(b)



$$\angle y = 27^\circ \text{ (isosceles } \triangle)$$

$$\begin{aligned}\angle x &= 180^\circ - 27^\circ - 27^\circ \\ &= 126^\circ\end{aligned}$$

(c)



$$\angle p = 110^\circ \text{ (corresponding angle with } 110^\circ \text{ angle given)}$$

$$\begin{aligned}\angle y &= 180^\circ - 110^\circ \\ &= 70^\circ \text{ (supplementary angle with } \angle p)\end{aligned}$$

$$\angle q = 110^\circ \text{ (supplementary angle with } \angle y)$$

$$\angle z = 110^\circ \text{ (corresponding angle with } \angle q)$$

$$\begin{aligned}\angle x &= 360^\circ - 110^\circ - 110^\circ - 70^\circ \\ &= 70^\circ\end{aligned}$$